

Benchmarking the beamprofiler M² Analysis Pipeline

Monte Carlo simulation study of systematic errors

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1 Motivation

The BEAMPROFILER application measures laser beam widths and M^2 beam quality factors using the ISO 11146-1 second-moment (D4 σ) method. Before applying the software to real measurements, it is necessary to quantify how accurately the pipeline recovers the true beam parameters under realistic noise and background conditions, and how those errors propagate into the final M^2 fit.

Two camera types are of particular interest:

- **Linear camera (Pyrocam-style):** detected signal proportional to intensity, $S \propto I$.
- **TPA camera (RayCi-style):** detection via two-photon absorption in silicon, so $S \propto I^2$. Used here for a 1750 nm beam, which is below the silicon single-photon bandgap.

Three questions drive the simulation:

1. How does the choice of background subtraction method affect the single-image width estimate at the beam waist?
2. How does the width estimation error vary across a z-scan, and why does this bias the M^2 fit even when the waist measurement is accurate?
3. How do the linear and TPA cameras compare, and what is the specific effect of discrete exposure compensation on TPA measurements?

2 Simulation design

2.1 Beam model

A simple astigmatic Gaussian beam is propagated analytically using the ISO 11146-1 free-space propagation law. The beam has independent waist radii and quality factors along x and y :

$$w_x(z) = w_{0x} \sqrt{1 + \left(\frac{z - z_{0x}}{z_{R,x}} \right)^2}, \quad z_{R,x} = \frac{\pi w_{0x}^2}{M_x^2 \lambda} \quad (1)$$

with true parameters $M_x^2 = 1.3$, $M_y^2 = 1.6$, $w_{0y}/w_{0x} = 0.8$, and an astigmatic separation of $z_{0y} - z_{0x} = 2 z_R$. The wavelength is $\lambda = 1750$ nm and the pixel pitch $\Delta x = 5.0$ μ m on a 256×256 pixel, 8-bit sensor.

2.2 Camera models

For a linear camera the raw ADC image at position z is

$$S_{\text{linear}}(x, y) = S_{\text{peak}} I_{\text{norm}}(x, y) + B + \varepsilon(x, y), \quad (2)$$

and for a TPA camera

$$S_{\text{TPA}}(x, y) = S_{\text{peak}} I_{\text{norm}}(x, y)^2 + B + \varepsilon(x, y), \quad (3)$$

where I_{norm} is the normalised (max = 1) Gaussian intensity profile, B is the background (dark current), and $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$ is additive noise with

$$\sigma_n = f_{\text{shot}} \sqrt{B} + \sigma_{\text{read}}, \quad f_{\text{shot}} = 0.05, \quad \sigma_{\text{read}} = 0.5 \text{ ADU}. \quad (4)$$

2.3 Discrete exposure model

In a real M^2 scan the beam intensity at the far-field positions is weaker than at the waist (geometrically by a factor of $w_{0x}w_{0y}/(w_xw_y)$, or its square for the TPA camera). In practice the operator compensates by increasing the camera exposure time. Camera exposure is available only in discrete doubling steps $\{1, 2, 4, \dots, 128\} \times$ the base exposure.

At each z -position the exposure step is chosen to bring the peak signal as close as possible to the target level $S_{\text{peak}} = 0.6 \times \text{ADC}_{\text{max}}$ without saturating. Background and noise both scale linearly with exposure time:

$$B(z) = B_0 \cdot t_{\text{exp}}(z), \quad S_{\text{peak}}(z) = S_{\text{peak},0} \cdot g(z) \cdot t_{\text{exp}}(z), \quad (5)$$

where $g(z) = w_{0x}w_{0y}/(w_x(z)w_y(z))$ is the geometric intensity factor (or g^2 for TPA). The background fraction $f_B = B_0/S_{\text{peak},0}$ is defined at unit exposure.

2.4 Background subtraction methods

Three methods are compared:

1. **Off** — no background subtraction; negative pixel values (from noise) are clamped to zero.
2. **Corner estimate** (ISO/TR 11146-3 §3.4.3) — the background mean is estimated from the four corner patches of the image (default patch size: 3.5% of the shorter sensor dimension). Pixels below $3\sigma_B$ after subtraction are zeroed.
3. **ISO statistical** (ISO/TR 11146-3 §3.4.2) — a uniform filter of size 3% of the sensor dimension is applied; pixels whose local mean exceeds $B_{\text{offset}} + 3\sigma_B/\sqrt{k^2}$ are classified as illuminated; the background is estimated from the remaining pixels.

For the TPA camera the background subtraction is performed in the raw ADC domain (before the $\sqrt{\cdot}$ transform), which is the physically correct order: the background is a linear additive offset in the ADC domain, and the $\sqrt{\cdot}$ transform then recovers the linear intensity.

2.5 Swept parameters

Parameter	Values	Description
f_B	1%, 2%, 5%, 10%	Background fraction at base exposure
w_0/r_{sensor}	0.10, 0.15, 0.20, 0.25	Beam waist / sensor half-width
Background mode	off, corner, ISO stat.	Subtraction method
Camera type	linear, TPA	Detection model

Table 1: Swept simulation parameters. All combinations are tested with $N = 15$ independent noise realisations per condition.

2.6 z-sampling

Z -positions follow the ISO 11146-1 recommendation with the following layout:

- 3 positions within $\pm 0.9 z_R$ of the mean waist (near-field),
- 2 positions between 2.1 and 2.5 z_R on the negative side (far-field),
- 3 positions between 2.1 and 3.0 z_R on the positive side (far-field).

The asymmetric far-field sampling (2 points on one side, 3 on the other) provides slightly more leverage on the divergence fit while keeping the maximum z extent at 3.0 z_R . At 3.0 z_R the beam

radius is $3.16 w_0$, so a beam filling 25% of the sensor at the waist will fill 79% at the farthest position — comfortably within the sensor.

2.7 Width and M^2 estimation

For each image the beam width is estimated in two ways:

1. **D4 σ** — ISO 11146-1 second-moment integration with iterative rotated-rectangle masking (mask factor = 3).
2. **Clip-level (13.5%)** — BeamGage-style marginal profile method: the width is the distance between the outermost crossings of the 13.5% ($= 1/e^2$) level of the marginal profile peak.

M^2 is obtained by fitting the hyperbolic propagation model to the D4 σ values across all z-positions:

$$d_{4\sigma}(z)^2 = A + B(z - z_0)^2, \quad M^2 = \frac{\pi}{4\lambda} \sqrt{A \cdot B}, \quad (6)$$

where $A = d_{4\sigma,0}^2$, B is the divergence term, and z_0 is the fitted waist position. All z-values are centred before fitting to avoid numerical issues when the absolute waist position is large.

3 Results and discussion

3.1 Single-image width accuracy at the beam waist

Figure 2 shows the mean relative width error $\Delta w/w = (\hat{w} - w_{\text{true}})/w_{\text{true}}$ for D4 σ_x at the beam waist across all three background methods, as a function of background fraction (1–10%) and beam/sensor ratio (10–25%). Figure 3 shows all four width metrics (D4 σ_x , D4 σ_y , clip $_x$, clip $_y$) for the ISO statistical method only.

BG off overestimates widths for the linear camera (background noise inflates the second-moment integral). For the TPA camera, BG off gives near-zero error when auto-ROI is applied: the squared intensity profile concentrates signal at the beam centre, so the auto-ROI captures almost purely beam signal and background noise within the crop has little effect on the second moments. Without auto-ROI, BG off would produce large errors for both cameras.

Corner method consistently underestimates widths for both cameras. Within the tested range ($w_0/r_{\text{sensor}} \leq 0.25$) the bias is approximately -3 to -5% for the linear camera and -13 to -16% for the TPA camera. The TPA underestimate arises from the $\sqrt{\cdot}$ transform compressing the beam wings non-uniformly relative to background residuals, not from the background estimation itself.

ISO statistical method gives slightly smaller errors than the corner method for the linear camera (-2 to -3%) and similar errors for the TPA camera (-13 to -20%). It shows larger variance for small beams because the 2D convolution kernel can misclassify some beam pixels as background when the beam covers only a small fraction of the sensor.

3.2 Z-dependent width error and its effect on M^2

The critical finding is that the width estimation error is *not constant* across the z-scan. At the beam waist the error is small for all methods, but it grows at far-field positions as the beam fills more of the sensor.

For the corner method the error growth is directly correlated with the beam/sensor filling fraction $w_x(z)/r_{\text{sensor}}$: as this fraction grows, beam signal contaminates the corner patches, the background is overestimated, and the beam width is underestimated. Since far-field widths are

underestimated more than near-field widths, the fitted hyperbola is flatter than the true one — leading to an underestimate of B and therefore of M^2 .

For the TPA camera an additional effect is visible: the width error curve shows discrete kinks that coincide exactly with the exposure step changes. Each time the exposure doubles, the background level doubles, the corner estimate shifts discontinuously, and the width bias jumps. These kinks appear as structured noise on the hyperbolic curve, degrading the M^2 fit beyond what random noise alone would cause.

The ISO statistical method is substantially more robust to both effects: its z-dependent error is flatter, because it identifies non-illuminated pixels globally rather than relying on the corners being unilluminated.

3.3 M^2 fit accuracy

The heatmap in Figure 4 shows the mean relative M^2 error $\Delta M^2/M_{\text{true}}^2$ across all conditions (beam/sensor 10–25%, background fraction 1–10%).

BG off is consistently the worst method. For the linear camera it strongly overestimates M^2 at small beam sizes (the inflated far-field widths push up the divergence term B), while for the TPA camera the behaviour is more variable. BG off should not be used for quantitative M^2 measurements.

Corner method gives moderate accuracy for the linear camera ($< 15\%$ error across the tested range) but is substantially worse for the TPA camera, with errors of -30 to -40% across all conditions. The large TPA error is primarily caused by the z-dependent bias described above: the far-field images have much lower signal-to-background ratio (because the TPA signal scales as geom^2 while the background scales only as geom), making the corner estimate unreliable at the positions that most influence the divergence fit.

ISO statistical method is the most accurate overall. For the linear camera errors remain below 10% across the tested conditions. For the TPA camera errors are -20 to -30% — smaller than the corner method but still significant, for the same physical reason.

3.4 Effect of ROI strategy on the corner background method

Figure 1 compares three variants of the corner background estimation method that differ in how the region of interest (ROI) is applied. This test uses a wider beam/sensor range (10–75%) than the main simulation to reveal the breakdown behaviour of each strategy:

Full sensor (§3.4.3 pure) No auto-ROI is applied. Background is estimated from the four corners of the full sensor image and the second-moment integral is computed over the full sensor. This is the method as literally described in ISO/TR 11146-3 §3.4.3.

Auto ROI + full-sensor BG The background is estimated from the corners of the full sensor (ensuring the corners contain only background), but the second-moment integral is restricted to an auto-ROI crop centred on the beam. This is the default behaviour of the BEAMPROFILER application.

Auto ROI + crop BG Both background estimation and second-moment integration use the auto-ROI crop. The background is estimated from the corners of the crop rather than the full sensor.

The ISO statistical method (§3.4.2) is included as a reference computed from the same images.

Findings

Full sensor (pure §3.4.3) performs well for small beams ($w_0/r_{\text{sensor}} \leq 0.25$) because the beam is well away from the sensor corners and the background estimate is accurate, giving errors of approximately -5 to -8% for the linear camera. However, the error grows above $w_0/r_{\text{sensor}} \approx 0.5$ as the beam wings reach the corners, contaminating the background estimate and causing systematic underestimation of the beam width. This directly confirms the §3.4.3 advisory that the method is reliable only when the beam width does not exceed half the sensor dimension.

Auto ROI + full-sensor BG gives results that are nearly identical to the pure §3.4.3 full-sensor method for the linear camera across the full range tested (10–75%), and also for the TPA camera at large beam sizes. For small beams the ISO statistical method is somewhat more accurate. The key advantage of this strategy is its *stability*: the error is nearly flat from 10% to 75% beam filling, with a systematic underestimate of approximately -8 to -9% for the linear camera, whereas the pure §3.4.3 method drops above 50%. This makes it the safer default choice when the beam/sensor ratio is not known in advance.

Auto ROI + crop BG performs worst at all beam sizes. A tight auto-ROI crop (padded to approximately 3σ per side) places the crop edges close enough to the beam that the corners of the crop still contain measurable beam signal. This leads to systematic overestimation of the background and consequently underestimates the beam width by 10–15% regardless of the true beam/sensor ratio.

For the TPA camera, all three corner strategies show a larger systematic underestimate (-10 to -15%) compared to the linear camera. This is not caused by the ROI strategy but is an intrinsic consequence of the $\sqrt{\cdot}$ transform applied after background subtraction, as discussed in §3.2.

Practical implication: the default BEAMPROFILER behaviour (auto ROI + full-sensor BG) is the most stable of the three corner strategies and does not require knowledge of the beam size relative to the sensor. The pure §3.4.3 full-sensor method is only preferable for very small beams ($< 25\%$ of sensor half-width); for larger its performance is similar to the previous strategy. Using the corners of the cropped image for background estimation should be avoided in all cases.

3.5 Practical recommendations

1. **Always use background subtraction.** BG off is unreliable regardless of background level.
2. **Use auto ROI with full-sensor background estimation** (the default app behaviour), or the ISO statistical method. Never estimate background from the corners of the cropped ROI.
3. **Prefer ISO statistical over corner method**, especially for TPA cameras and for beams filling $> 25\%$ of the sensor.
4. **Keep the beam waist below 25% of the sensor half-width** during an M^2 scan. At 25% waist filling, the farthest far-field position ($3.0 z_R$) fills 79% of the sensor — well within the sensor while providing good leverage on the divergence fit.
5. **For TPA cameras**, the discrete exposure stepping introduces structured errors on the propagation curve. Using finer exposure steps (e.g. $\times 1.5$ increments rather than $\times 2$) would reduce this effect.
6. **Use a dark frame** where possible. Subtracting a separately acquired dark image removes the systematic background offset more accurately than any estimation method.

4 Figures

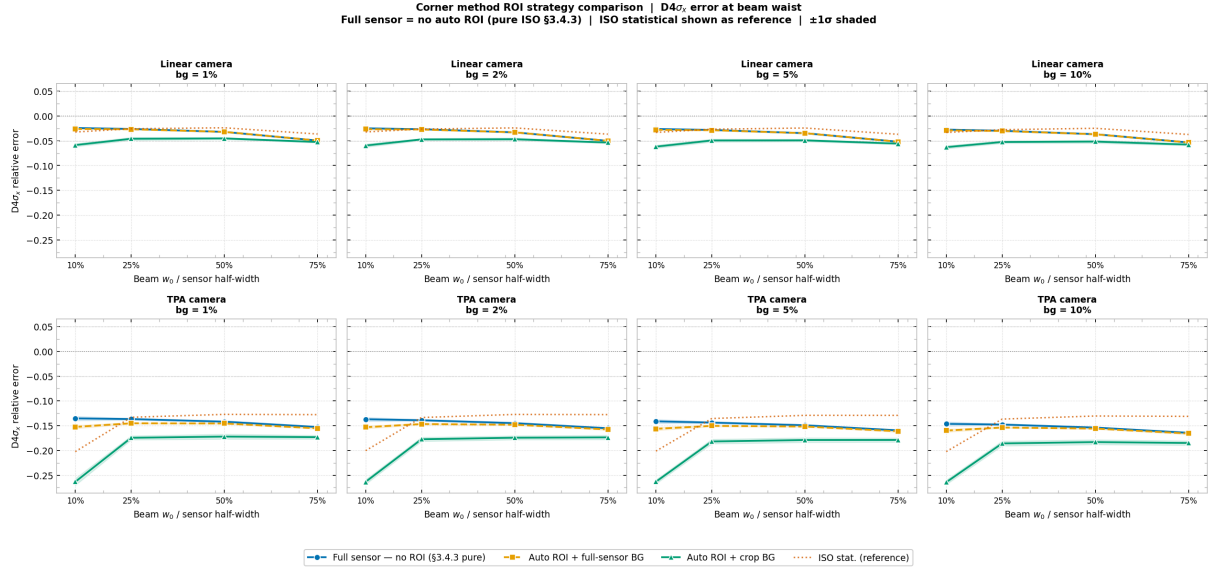


Figure 1: Comparison of three corner background estimation strategies for the $D4\sigma_x$ width error at the beam waist, as a function of beam/sensor ratio and background fraction. *Full sensor* applies no auto-ROI (pure ISO/TR 11146-3 §3.4.3). *Auto ROI + full-sensor BG* restricts integration to an auto-ROI crop while estimating background from the full sensor corners (default BEAMPROFILER behaviour). *Auto ROI + crop BG* estimates background from the corners of the crop. The ISO statistical method (§3.4.2) is shown as a dotted reference line; it was computed from the same images. Shaded bands: $\pm 1\sigma$ over 50 noise realisations.

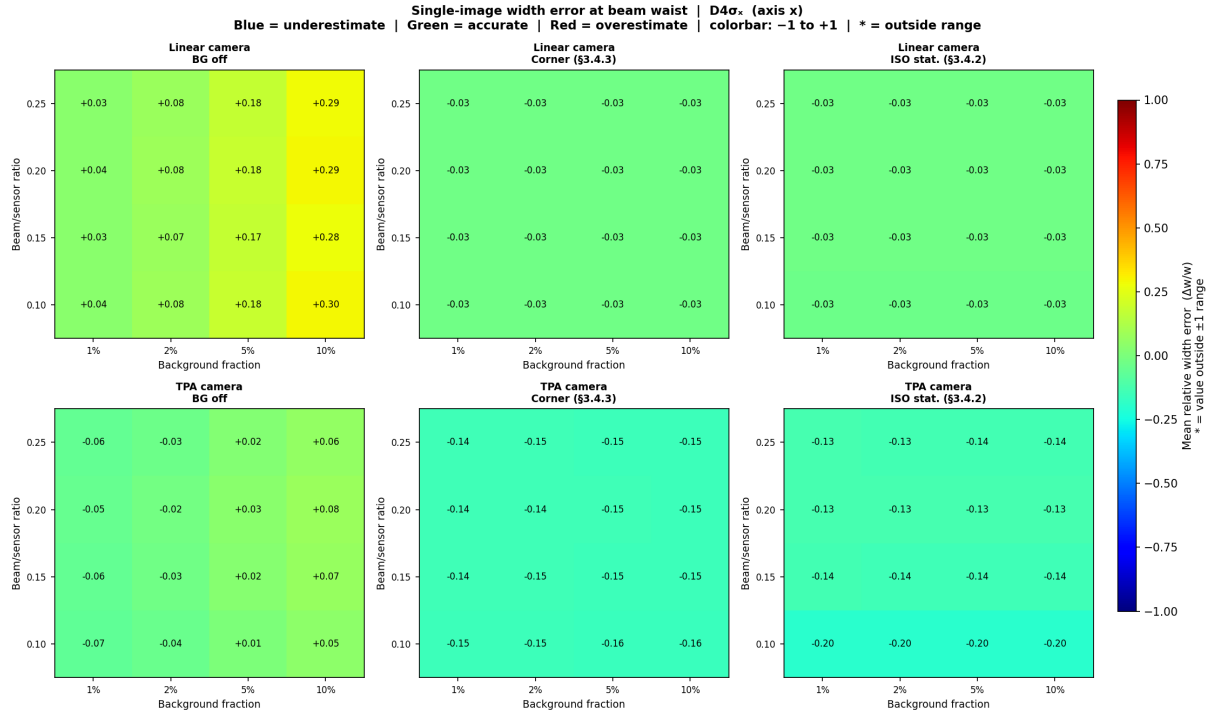


Figure 2: Mean relative $D4\sigma_x$ width error at the beam waist for all three background methods (BG off, corner §3.4.3, ISO statistical §3.4.2), both camera types (linear and TPA), and all swept conditions. Colorbar: jet, -1 to +1. Blue = underestimate, green = accurate, red = overestimate. Values outside ± 1 are marked with *.

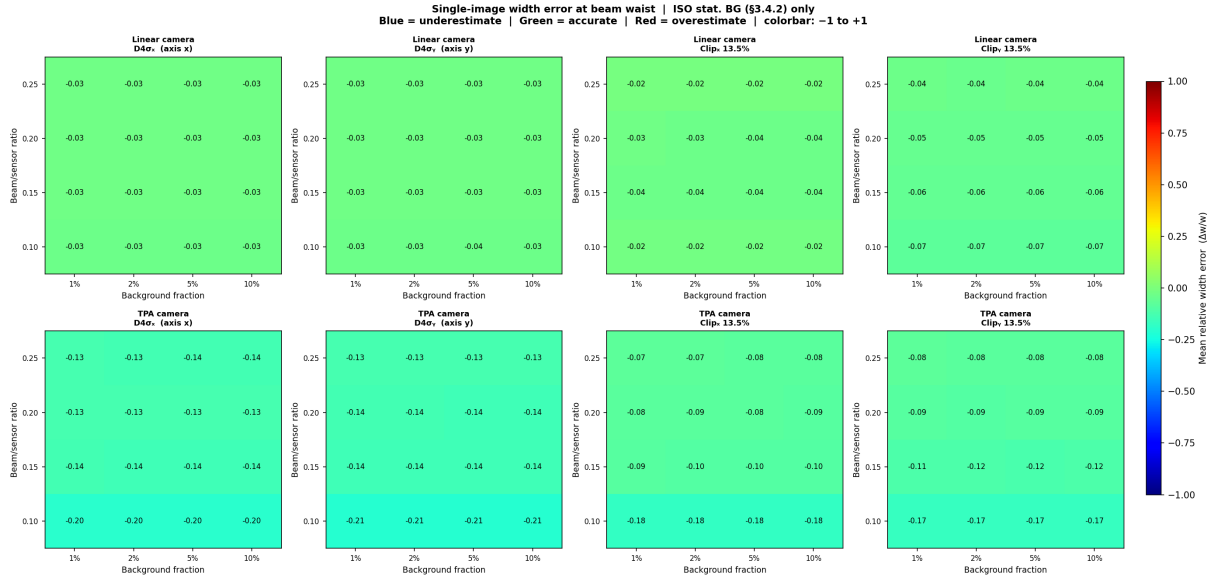


Figure 3: Mean relative width error $\Delta w_x/w_{x,\text{true}}$ at the beam waist for all four width metrics ($D4\sigma_x$, $D4\sigma_y$, clip_x , clip_y) using the ISO statistical background method. Colorbar: jet, -1 to +1. Blue = underestimate, green = accurate, red = overestimate. Values outside ± 1 are marked with *.

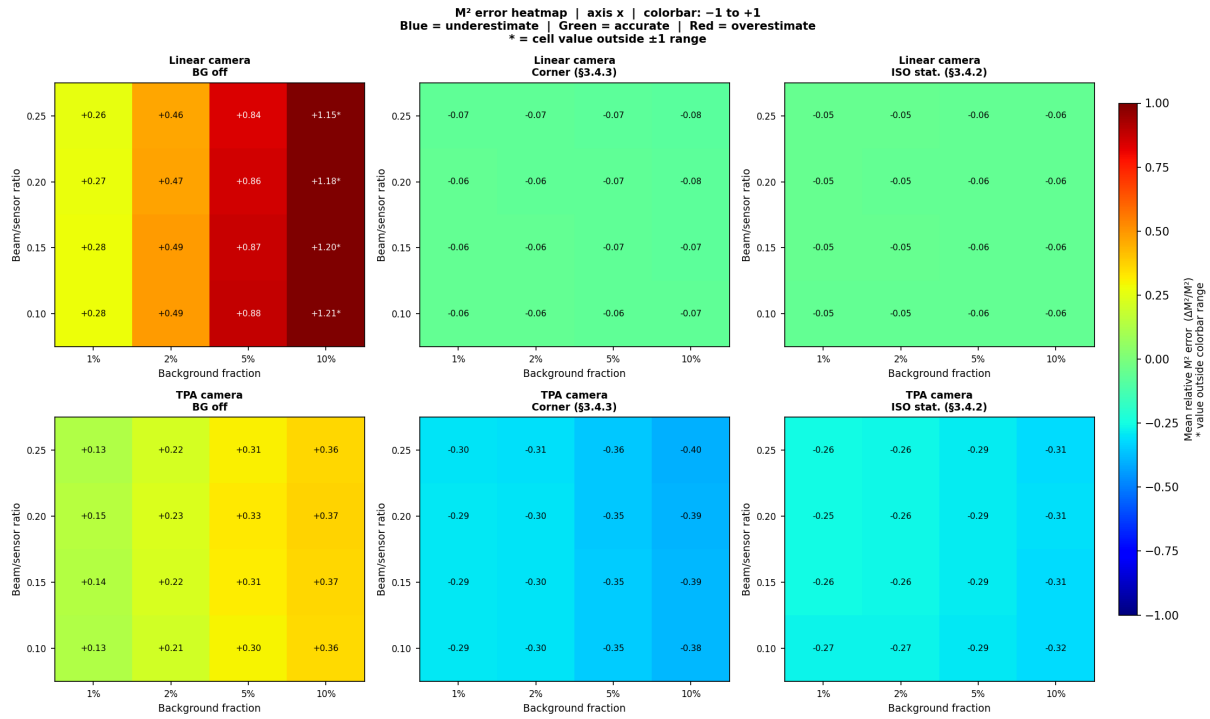


Figure 4: Mean relative M² error across all background subtraction methods. The ISO statistical method (right column) gives the smallest and most uniform errors. BG off (left column) is unreliable for all conditions. Values outside ± 1 are marked with *.

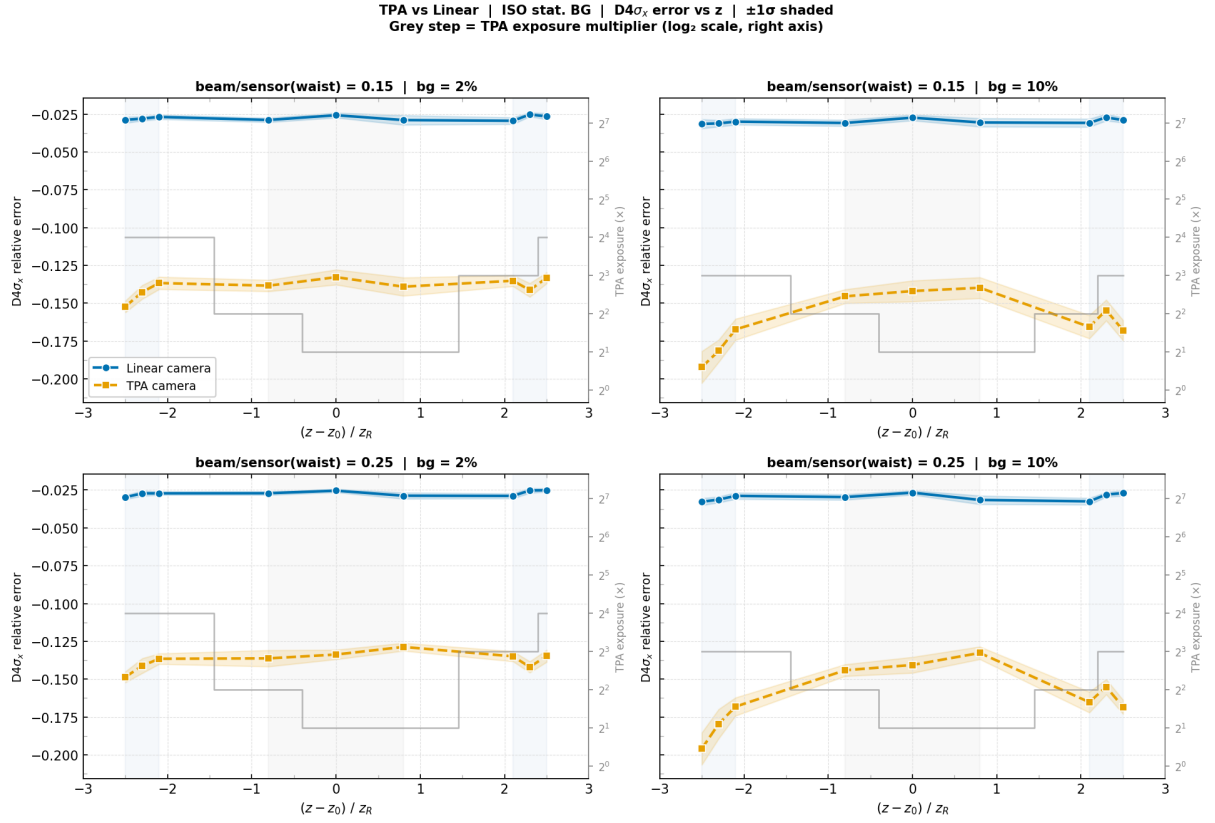


Figure 5: $D4\sigma_x$ relative error as a function of z-position (in units of z_R) for TPA and linear cameras with ISO statistical background subtraction. The grey step function (right axis, log₂ scale) shows the discrete TPA exposure multiplier. Jumps in the TPA error curve coincide exactly with exposure steps. Shaded regions: grey = near-waist zone, blue = far-field zone.